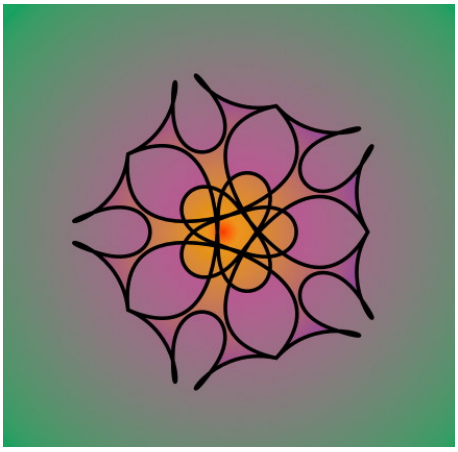

Group Theory

An Introduction to Algebraic Symmetry and Structures

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My Website

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> ”Symmetry, as wide or as narrow as you may define its meaning, is one idea by which man through the ages has tried to comprehend and create order, beauty, and perfection.”
>
> *Symmetry*
> HERMANN WEYL

1.1 Basic Definitions

The mathematical definition of a group is modeled after the properties of physical symmetries. When we study symmetries, we care about how they combine, invert, and leave certain aspects of a system invariant.

1.1.1 The Axioms of a Group

Definition 1.1 – Group

A group $(G, *)$ is a set G together with a well defined binary operation $* : G \times G \to G$ that satisfies the following four axioms:

1. **Closure:** $\forall a, b \in G$, the element $a * b \in G$.
2. **Associativity:** $\forall a, b, c \in G$, we have $(a * b) * c = a * (b * c)$.
3. **Identity:** $\exists e \in G$ such that for all $a \in G$, $e * a = a * e = a$.
4. **Inverse:** $\forall a \in G$, there exists an element $a^{-1} \in G$ such that $a * a^{-1} = a^{-1} * a = e$.

Lemma 1.1.

(a) ***Right cancellation law****: $\forall a, b, c \in G$, if $a * b = a * c$ then $b = c$.*

(b) ***Left cancellation law****: $\forall a, b, c \in G$, if $b * a = c * a$ then $b = c$.*

(c) The identity e of a group $(G, *)$ is unique.

(d) The inverse a^{-1} for an element $a \in G$ is unique.

Proof.
(a) Consider elements $a, b, c \in G$ such that $a * b = a * c$. Now, we operate a^{-1} to the right for both sides.

$$a^{-1} * (a * b) = a^{-1} * (a * c) \tag{1.1}$$
$$a^{-1} * a * b = a^{-1} * a * c \ (\because \text{Associativity of group elements}) \tag{1.2}$$
$$e * b = e * c \tag{1.3}$$
$$\therefore b = c \quad \square \tag{1.4}$$

(b) Consider elements $a, b, c \in G$ such that $b * a = c * a$. Now, we operate a^{-1} to the left for both sides.

$$(b * a) * a^{-1} = (c * a) * a^{-1} \tag{1.5}$$
$$b * a * a^{-1} = c * a * a^{-1} \ (\because \text{Associativity of group elements}) \tag{1.6}$$
$$b * e = c * e \tag{1.7}$$
$$\therefore b = c \quad \square \tag{1.8}$$

(c) Let e and e' be elements of G s.t. $\forall a \in G$, $e * a = a * e = a$ and $e' * a = a * e' = a$. Then:

$$a = a * e = a * e' \tag{1.9}$$
$$e = e' \quad (\because \text{Right cancellation}) \tag{1.10}$$

Since $e = e'$, there is only one identity element, and the identity is unique. $\square$
(d) Let a' and a'' be two inverses of $a \in G$, then $a' * a = a * a' = e$ and $a'' * a = a * a'' = e$, where e is the identity. Then:

$$e = a' * a = a'' * a \tag{1.11}$$
$$a' = a'' \quad (\because \text{Right cancellation}) \tag{1.12}$$

Since $a' = a''$, there is only one inverse element, and the inverse is unique. $\square$

Example 1.1 – Some Groups

1. For $n \geq 3$, the group of symmetries of a regular n–gon is called the dihedral group, under the composition of symmetries, with the identity being the trivial symmetry. This group has n reflections and n rotations of angle $2k\pi/n$ where $k \in \mathbb{Z}$.
2. Additive groups like $(\mathbb{Z}, +), (\mathbb{Q}, +), (\mathbb{R}, +)$ and $(\mathbb{C}, +)$ have the identity 0 and the inverse of element a as $-a$.
3. $C_n = \left\{e^{i2\pi k/n} \,|\, 0 \leq k \leq n-1\right\}$, i.e. the set of complex n'th roots of 1, form a group under complex mutiplication.
4. Multiplicative groups like $(\mathbb{Q}^x, \cdot), (\mathbb{R}^x, \cdot)$ and $(\mathbb{C}^x, \cdot)$ have the identity 1 and

the inverse of element z being $1/z$, here $F^x = F \setminus \{0\}$.